

own a 3D printer, you can find all the printable parts at [reprap.org/wiki/Prusa_Mendel_\(iteration_2\)#Printed_Parts](http://reprap.org/wiki/Prusa_Mendel_(iteration_2)#Printed_Parts) and make all the parts yourself.

The RepRap Prusa Mendel 2 is what we are building as it is the easiest to put together and gives you the advantage of having a full sized 200mm x 200mm x 100mm work space. While the RepRap Prusa Mendel may be the easiest to build, it is still complicated and would require over 40 hours of work to actually build it. This article is intended to so present a detailed overview of all the components the Prusa Mendel contains and how they work. It would also help one familiarise themselves with how 3D printers work.

What should your shopping list contain?

While the entire list is easily available at reprap.org/wiki/Prusa_Mendel, the Prusa Mendel is made up of:

- 1) The Frame
- 2) The X, the Y and the Z axis with their rails and brushings
- 3) The Extruder
- 4) The X Carriage
- 5) The Print Plate
- 6) The Electronic Components
- 7) Stepper Motors and their controllers and
- 8) End Stops
- 9) Filaments
- 10) Programmes

The Frame:

The frame is comprised of two triangle shaped structures erected facing each other and joined with the same threaded rods that make the triangles. These parts are held together using printable parts. On this structure we will be mounting the X, Y and Z axes using even more printable parts.

Smooth rods that serve as rails for the Z axis will be installed vertically on the outside of the frame while the Y axis' rails will be perpendicular on the inside.



The Frame

The Axes

In the Prusa Mendel when viewed head on, the X axis moves left to right, the Y moves